

Name: _____

Recitation: _____

Math 340
Final Exam
December 17, 2025

Problem	Score	Problem	Score
1		8	
2		9	
3		10	
4		11	
5		12	
6		13	
7		14	
		Total	

All problems are worth 10 points. The exam is closed book, but you may use a calculator and one $8\frac{1}{2} \times 11$ " sheet of handwritten notes (both sides). You must show your work to receive full credit. Write solutions in explicit form if possible. All problems have a solution that can be found using the techniques of this class.

Pledge:

On my honor, as a student, I have neither given nor received unauthorized aid on this

examination: _____

(signature)

(date)

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Problem 1

Find the general solution to $\frac{dy}{dx} = (2x - 1)(y + 1)$

Separable

$$\frac{dy}{y+1} = (2x-1) dx$$

$$\int \frac{dy}{y+1} = \int (2x-1) dx$$

$$\ln|y+1| = x^2 - x + C$$

$$y+1 = ke^{x^2-x}$$

$$y = ke^{x^2-x} - 1$$

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Problem 2

Find the general solution to $y'' + 7y' + 12y = 17\cos(x)$.

Homo: $(D^2 + 7D + 12)y = 0$

roots $-3, -4$

$$y_h = C_1 e^{-3x} + C_2 e^{-4x}$$

Part Guess $y_p = A\cos(x) + B\sin(x)$

$$y_p' = -A\sin(x) + B\cos(x)$$

$$y_p'' = -A\cos(x) - B\sin(x)$$

$$y_p'' + 7y_p' + 12y_p = (A + 7B + 12A)\cos(x) + (-B - 7A + 12B)\sin(x)$$

$$= 11A + 7B \cos(x) + (-7A + 11B) \sin(x)$$

Set $\cos(x)$

$$11A + 7B = 17 \rightarrow 77A + 49B = 7 \cdot 17 (=109)$$

$$-7A + 11B = 0$$

$$170B = 7 \cdot 17$$

$$B = \frac{7}{10}$$

$$121A + 77B = 17 \cdot 11 (=187)$$

$$49A - 77B = 0$$

$$170A = 17 \cdot 11$$

$$A = \frac{11}{10}$$

$$y_p = 1.1\cos(x) + 0.7\sin(x)$$

$$y = 1.1\cos(x) + 0.7\sin(x) + C_1 e^{-3x} + C_2 e^{-4x}$$

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Problem 3

Solve the initial value problem

$$\frac{dy}{dx} = y - xy^2, y(0) = 1.$$

Bernoulli:

$$\frac{dy}{dx} - y = -xy^2$$

$$-V^{-2} \frac{dV}{dx} - V^{-1} = -xV^{-2}$$

$$\frac{dV}{dx} + V = x$$

$$\text{Let } y = V^{\frac{1}{1-2}} = V^{-1}$$

$$\frac{dy}{dx} = -V^{-2} \frac{dV}{dx}$$

Constant Coeff Linear

$$(D+1)V = x$$

$$V_h = C_1 e^{-x}$$

$$V_p = Ax + B$$

$$V_p' = A$$

$$V_p = x - 1$$

$$V = x - 1 + Ce^{-x}$$

$$y = \frac{1}{x - 1 + Ce^x}$$

$$Ax + B + A = x$$

$$A = 1 \Rightarrow B = -1$$

$$B + A = 0$$

$$y(0) = \frac{1}{C-1} = 1$$

$$C = 2$$

$$y = \frac{1}{x - 1 + 2e^{-x}}$$

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Problem 4

Solve the initial value problem $x'' + 6x' + 25x = \delta(t)$, $x(0)=0$, $x'(0)=0$.

Take Laplace transform

$$(s^2 + 6s + 25)X = 1$$

$$X = \frac{1}{s^2 + 6s + 25}$$

$$= \frac{1}{s^2 + 6s + 9 + 16}$$

$$= \frac{1}{(s+3)^2 + 4^2}$$

$$\mathcal{L}\{e^{-3t} \sin(4t)\}$$

$$= \frac{4}{(s+3)^2 + 4^2}$$

$$\text{So } \boxed{x = \frac{1}{4} e^{-3t} \sin(4t)}$$

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Problem 5

Solve the initial value problem. Note that your answer will have an a in it.

$$x'' + 10x' + 16x = 0, x(0) = a, x'(0) = 0$$

Chapter 2

$$(D^2 + 10D + 16)x = 0$$

roots $-2, -8$

$$x = C_1 e^{-2x} + C_2 e^{-8x}$$

$$x' = -2C_1 e^{-2x} - 8C_2 e^{-8x}$$

$$x(0) = C_1 + C_2 = a$$

$$x'(0) = -2C_1 - 8C_2 = 0$$

$$\begin{array}{r} 2C_1 + 2C_2 = 2a \\ -2C_1 - 8C_2 = 0 \end{array}$$

$$-6C_2 = 2a$$

$$C_2 = -\frac{1}{3}a$$

$$C_1 + C_2 = a$$

$$C_1 - \frac{1}{3}a = a$$

$$C_1 = \frac{4}{3}a$$

$$x = \frac{4a}{3}e^{-2x} - \frac{a}{3}e^{-8x}$$

Chapter 3

$$s^2 \bar{x} - as + 10(s\bar{x} - a) + 16\bar{x} = 0$$

$$(s^2 + 10s + 16)\bar{x} = as + 10a$$

$$\bar{x} = \frac{as + 10a}{s^2 + 10s + 16}$$

$$= \frac{as + 10a}{(s+2)(s+8)}$$

$$= A \frac{1}{s+2} + B \frac{1}{s+8}$$

$$as + 10a = A(s+8) + B(s+2)$$

$$s = -2 \quad 8a = 6A$$

$$A = \frac{8}{6}a = \frac{4}{3}a$$

$$s = -8 \quad 2a = -6B$$

$$-\frac{1}{3}a = B$$

$$x = \frac{4a}{3}e^{-2x} - \frac{a}{3}e^{-8x}$$

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Problem 6

A mass of 5kg is attached to a spring, causing it to stretch 0.1 meters (10 cm). What is critical damping for this spring-mass system. Use $g = 9.8$ m/sec.

$$mg = k\Delta l$$

$$5 \cdot 9.8 = k \cdot 0.1$$

$$490 \text{ kg/sec}^2 = k$$

Critical Damping when

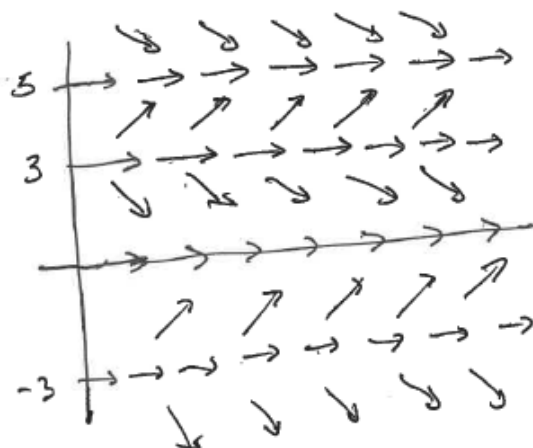
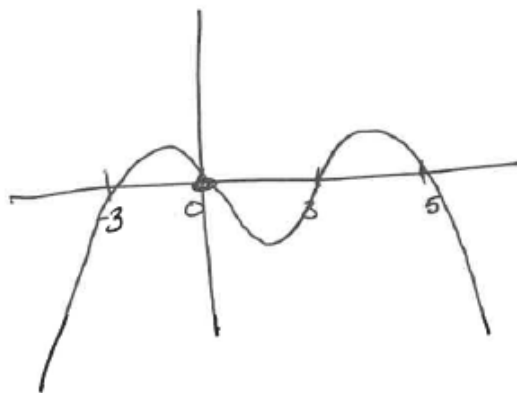
$$C = \sqrt{4km} = \sqrt{9800} \\ = 98.99 \text{ kg/sec}$$

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Problem 7

Find and classify the equilibria for $\frac{dx}{dt} = x(x^2 - 9)(5 - x)$.

roots are at $0, \pm 3, 5$ $-x^4$ leading term so
going down
as $x \rightarrow \pm \infty$



$x = 0, 5$ are
stable equilibria

$x = -3, 3$ are
unstable equilibria

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Problem 8

Solve the initial value problem

$$\frac{dx}{dt} = x - y, \quad x(0) = 0$$

$$\frac{dy}{dt} = 2x + 4y, \quad y(0) = 3$$

Take

$$sX = X - Y$$

$$sY - 3 = 2X + 4Y$$

$$(s-1)X + Y = 0$$

$$-2X + (s-4)Y = 3$$

$$(s-4)(s-1)X + (s-4)Y = 0$$

$$2X - (s-4)Y = -3$$

$$[(s-4)(s-1)+2]X = -3$$

$$X = \frac{-3}{s^2-5s+6}$$

$$\frac{-3}{s^2-5s+6} = A \frac{1}{s-2} + B \frac{1}{s-3}$$

$$-3 = A(s-3) + B(s-2)$$

$$s=3 \quad -3 = B$$

$$s=2 \quad -3 = -A$$

$$X = 3e^{2t} - 3e^{3t}$$

$$2(s-1) + 2Y = 0$$

$$-2(s-1) + (s-4)(s-1)Y = 3s-3$$

$$(s^2-5s+4+2)Y = 3s-3$$

$$Y = \frac{3s-3}{s^2-5s+6}$$

$$\frac{3s-3}{s^2-5s+6} = A \frac{1}{s-2} + B \frac{1}{s-3}$$

$$3s-3 = A(s-3) + B(s-2)$$

$$s=2 \quad 3 = -A \rightarrow A = -3$$

$$s=3 \quad 6 = B$$

$$Y(t) = -3e^{2t} + 6e^{3t}$$

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Problem 9

Solve the boundary value problem:

$$x'' + 2x' - 15x = 0 \text{ with } x(0)=2 \text{ and } \lim_{t \rightarrow \infty} x(t) = 0.$$

$$(D^2 + 2D - 15)x = 0$$

$$(D+5)(D-3)$$

roots -5, 3

$$x = C_1 e^{-5t} + C_2 e^{3t}$$

$$x(0) = C_1 + C_2 \stackrel{\text{Set}}{=} 2$$

$$\lim_{t \rightarrow \infty} x(t) = C_2 \cdot \infty \stackrel{\text{Set}}{=} 0 \Rightarrow C_2 = 0$$

$$\text{So } C_1 = 2$$

$$\boxed{x = 2e^{-5t}}$$

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Problem 10

A comet has perihelion at 0.4 AU and aphelion at 5 AU. What is the period of the comet (in years).

$$\text{Semi-Major Axis} = \frac{0.4 + 5}{2} = 2.7$$

$$T^2 = a^3 \quad (\text{since use of A.U. and years makes constant} = 1)$$

$$T = (2.7)^{3/2} \approx 4.4 \text{ years}$$

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Problem 11

What is the a priori lower bound for the radius of convergence of the series solution about $x_0 = 0$ for

$$(x^2 + 4)y'' + (x - 1)y' + 2y = 0, y(0) = 1, y'(0) = 2.$$

Singular Points are at $\pm 2i$
R.O.C. ≥ 2

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Problem 12

Find the general solution to

$$x^2 y'' + 7xy' + 7y = 0.$$

Euler Equation

$$\begin{aligned}\text{Try } y &= x^r \\ y' &= r x^{r-1} \\ y'' &= r(r-1) x^{r-2}\end{aligned}$$

$$\begin{aligned}x^2 y'' + 7xy' + 7y &= 0 \\ r(r-1)x^r + 7rx^r + 7x^r &= 0 \\ (r^2 + 6r + 7)x^r &= 0\end{aligned}$$

$$\text{roots } \frac{-6 \pm \sqrt{36 - 28}}{2} = \frac{-6 \pm \sqrt{8}}{2} = -3 \pm \sqrt{2}$$

$$y = C_1 x^{(-3+\sqrt{2})} + C_2 x^{(-3-\sqrt{2})}$$

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Problem 13

A two-tank CSTR consists of two 2000 liter tanks. Solution flows into the first tank, from the first tank to the second tank, and out of the second tank all at a rate of 20 liters/min. Suppose 500 grams of a contaminant is introduced to the first tank. What is the maximum concentration reached for the contaminant in the second tank?

Let x_1 = contaminant in tank 1
 x_2 = contaminant in tank 2

Our system of equations is

$$\begin{aligned}\frac{dx_1}{dt} &= -\frac{x_1}{100} & x_1(0) &= 500 & \text{(No new contaminant flowing in, only out)} \\ \frac{dx_2}{dt} &= \frac{x_1}{100} - \frac{x_2}{100} & x_2(0) &= 0 & \text{(Flowing in from tank 1 and out of tank 2)}\end{aligned}$$

Take $\mathcal{L}$

$$s\bar{x}_1 - 500 - \frac{x_1}{100} \rightarrow \bar{x}_1 = \frac{500}{s + \frac{1}{100}}$$

$$s\bar{x}_2 = \frac{1}{100}\bar{x}_1 - \frac{1}{100}\bar{x}_2 \rightarrow (s + \frac{1}{100})\bar{x}_2 = \frac{5}{s + \frac{1}{100}}$$

$$\bar{x}_2 = \frac{5}{(s + \frac{1}{100})^2} \Rightarrow x_2 = 5te^{-t/100}$$

Max contaminant in tank 2 when $x_2' = 0$

$$5e^{-t/100} - \frac{5t}{100}e^{-t/100} = 0 \rightarrow t = 100$$

$$\text{Contaminant} = 500e^{-1}$$

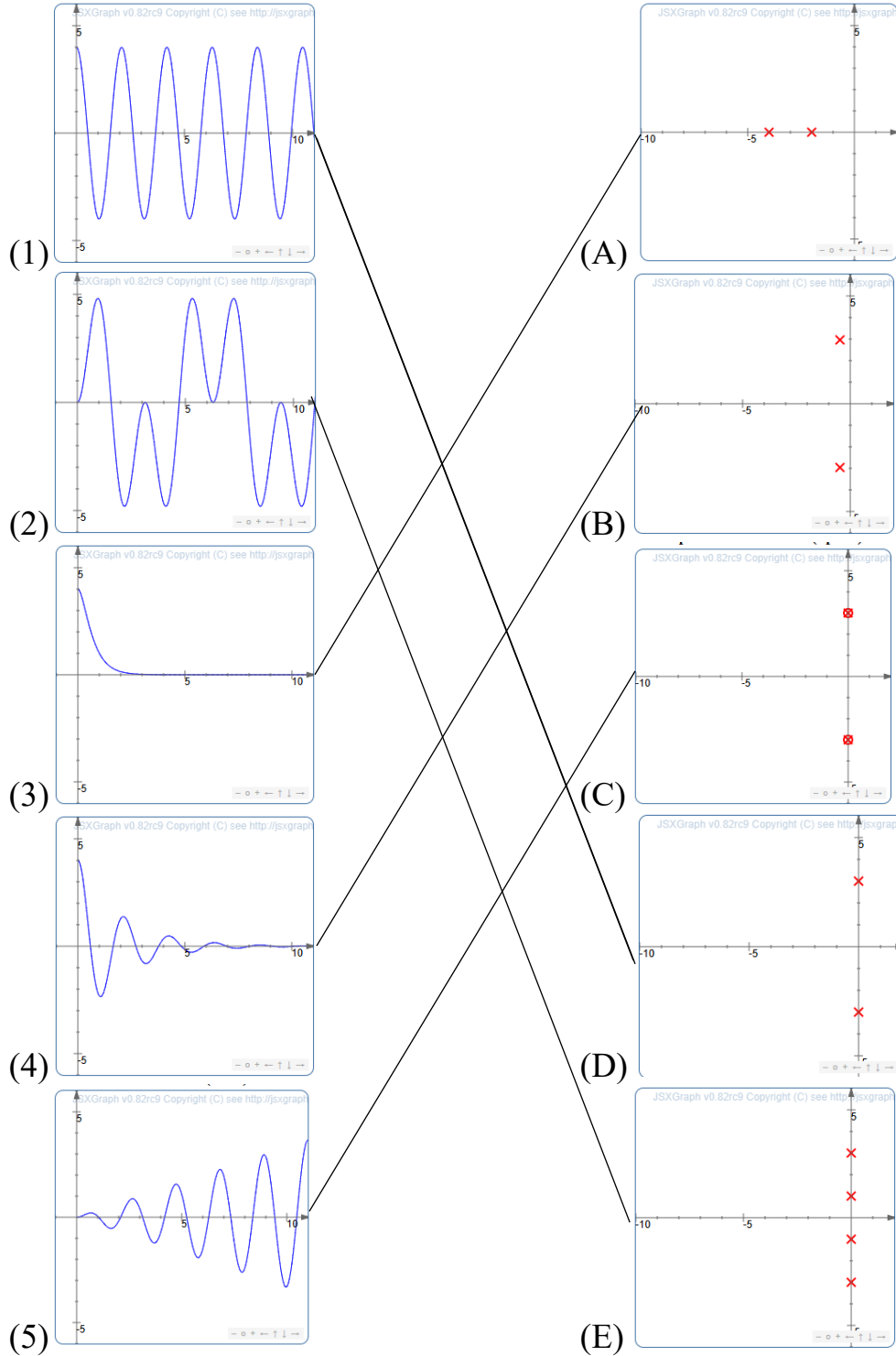
$$\text{Concentration} = \frac{500e^{-1}}{2000} = \boxed{\frac{1}{4e} \text{ g/l}}$$

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Problem 14

Match the following graphs of a function vs. time with the corresponding graph showing the location of the poles of the Laplace transform.

1D, 2E, 3A, 4B, 5C



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